DATA ANALYSIS PERSONAL PROJECT

Laptop Market Size Analysis in India

Using 1.001 product listings from the Indian laptop market, this project maps brand concentration, pricing patterns, and hardware trends to identify where real value sits and where premium pricing is driven more by brand than by specs.

2026

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01 Project Overview

Objectives · Tools · Methodology

Project Overview — Objectives

Why This Analysis?

To analyze the structure of the Indian laptop market, identify pricing and performance trends, and uncover value-for-money opportunities across different brands.

Market Structure

Map brand presence, concentration, and product diversity across the Indian laptop market.

Pricing & Trends

Identify price distribution patterns, segment boundaries, and premium vs. budget dynamics.

Value for Money

Surface brands and models delivering the best specs-to-price ratio for consumers.

Project Overview — Tools & Methodology

Microsoft Excel

- Typo correction & standardization
- Remove irrelevant columns
- Unmatched value fixes

Python — Pandas

- Duplicate removal
- Missing value imputation
- RAM / Storage standardization

Python — NumPy

- Statistical computations
- Specs score aggregation
- VFM score calculation

Matplotlib & Seaborn

- Bar, pie, histogram charts
- Scatter & boxplot visuals
- Publication-ready figures

Raw Data → Excel Cleaning → Python EDA → Visualization → Insights

02 Dataset Overview

Raw Data Profile · Source · Variables

Dataset Overview — Raw Data Profile

1.011	Raw Records
1.001	Cleaned Records
25	Brands
16	Variables
₹9.999	Min Price
₹617.990	Max Price

Dataset Metadata

Source	Kaggle: Indian Laptop Market Dataset
License	CC BY-SA 4.0: allows users to distribute, combine, adapt, and build upon the material in any medium or format, provided they give credit (Attribution) to the original creator and release derivative works under the same license
Format	CSV (Semicolon-delimited in raw; Comma-delimited after cleaning)
Time Scope	Static snapshot of product listings (2025)

Dataset Overview — Key Variables

Variable	Type	Description
name	String	Full product name of the laptop
brand	String	Manufacturer / brand name
price	Numeric	Retail price in Indian Rupees (₹)
ratings	Float	Average user rating (1–5 scale)
specs_score	Float	Composite hardware specification score
os	String	Operating system
processor	String	CPU model and generation
ram_size	String	RAM capacity (e.g., 8 GB, 16 GB)
ram_type	String	RAM technology (DDR4, DDR5, LPDDR5, etc.)
storage	String	Storage capacity and type (SSD / eMMC)
graphics	String	GPU model (integrated or dedicated)
screen_size	Float	Display size in inches
resoultion	String	Screen resolution (e.g., 1920×1080)
cores	String	CPU core count and configuration
threads	Integer	CPU thread count
warranty_months	Float	Warranty period in months (cleaned from text)

03 Data Cleaning

Excel · Python

Data Cleaning — Excel & Python Steps

Cleaning with Excel

Typo Correction

Fixed brand/OS misspellings (e.g., "Widnows" → "Windows"); standardized product names.

Unmatched Values

Resolved inconsistent categorical entries; aligned values to canonical labels.

Column Removal

Dropped "image_url" column — not relevant for quantitative analysis.

Format Standardisation

Ensured consistent delimiter (CSV) and encoding before Python ingestion.

Cleaning with Python

Duplicate Removal

Identified and dropped 10 duplicate records, reducing dataset from 1.011 → 1.001.

Missing Value Imputation

Filled null values using mode/median for categorical and numeric fields.

RAM Standardization

Parsed "16 GB DDR5 RAM" → ram_size="16 GB", ram_type="DDR5".

Storage Standardization

Separated storage capacity and type into structured fields.

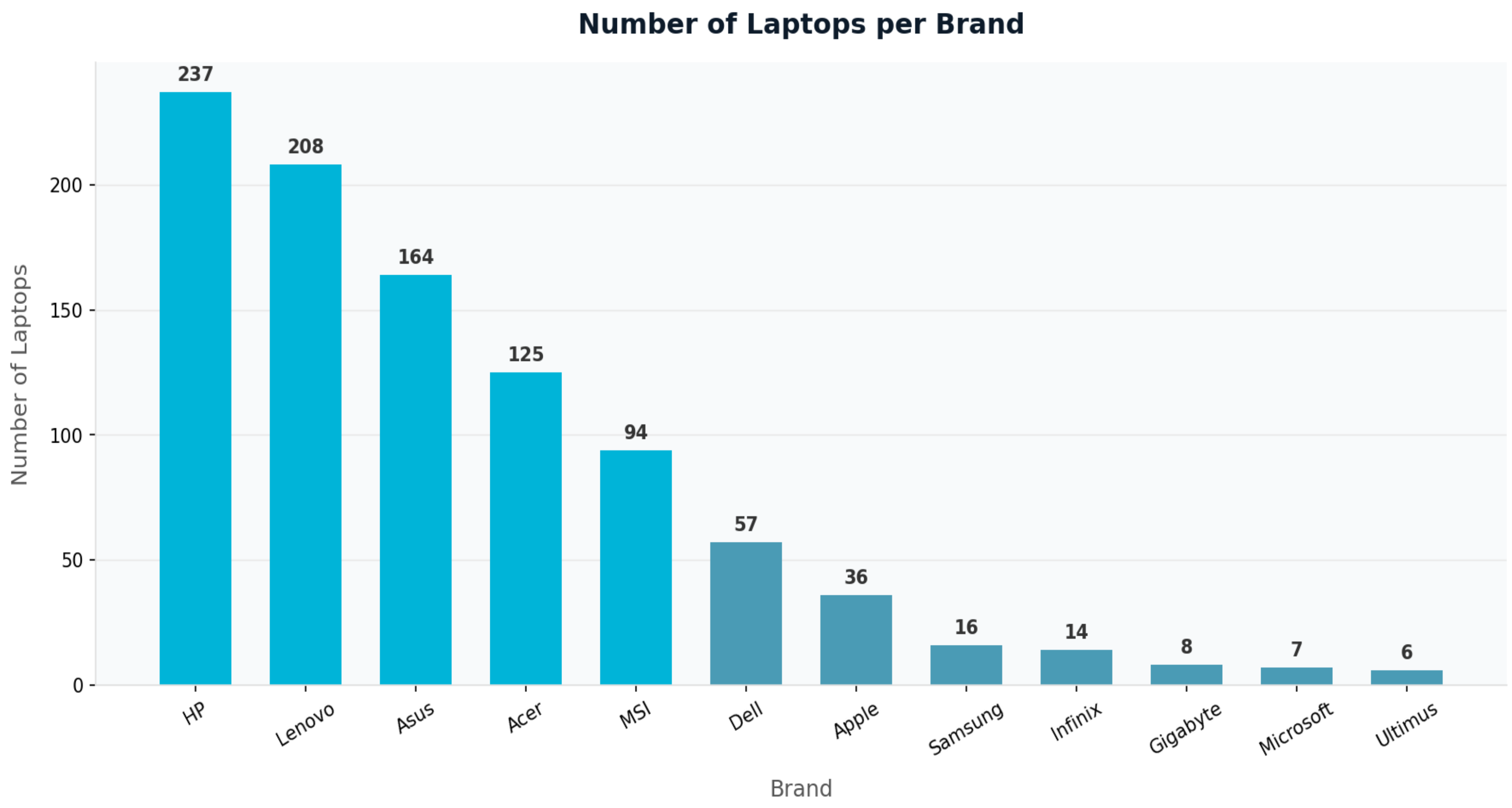
Type Conversion

Converted price, specs_score to numeric; warranty string → warranty_months (float).

04 Market Overview

Brand Distribution · Market Share

Market Overview — Laptops per Brand



Quick Stats

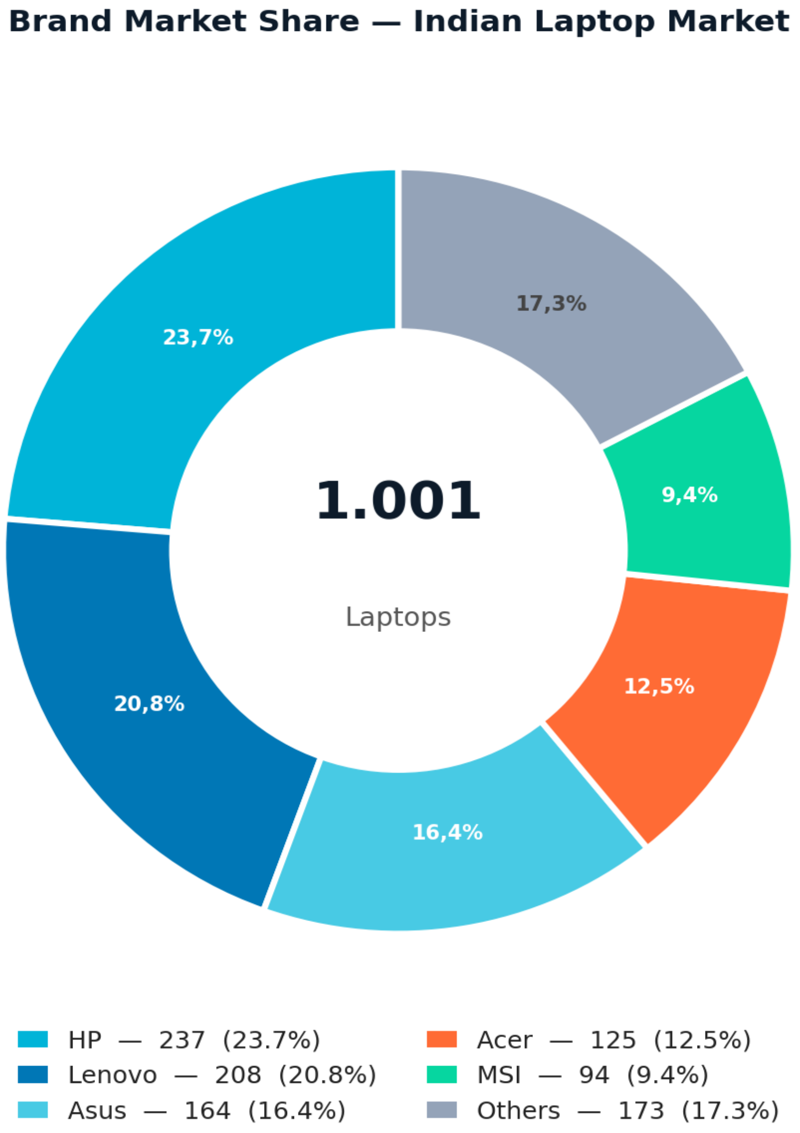
25
Brands listed

237
HP: Top Brand

82.7%
Top 5 share

5
Dominant brands

Market Overview — Brand Market Share



Market Concentration Analysis

HP leads the pack

Holds 23.7% of the market with 237 listed models, giving it the broadest product catalogue of any brand in the Indian market.

Lenovo is right behind

Sits at 20.8%, close enough to HP that the gap between 1st and 2nd is effectively a tie in catalogue terms.

Five Brands, most of the market

HP, Lenovo, Asus, Acer, and MSI account for 82.7% of all product listings. The remaining 20 brands share less than one-fifth of the market between them.

Concentration is high at the top

HP, Lenovo, and Asus hold more than 60% of all listings. In most industries, that level of concentration would be considered a strong oligopoly.

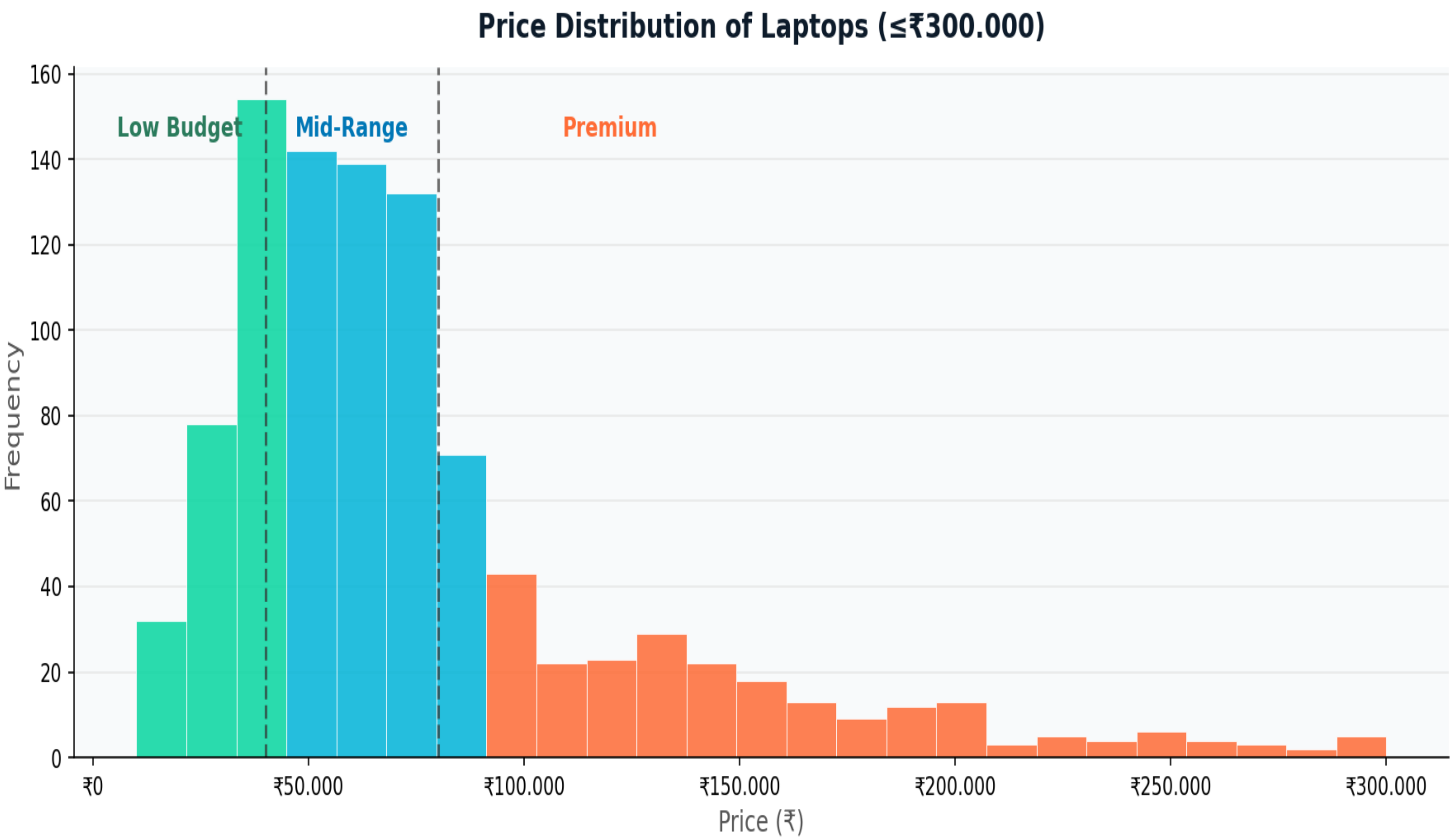
Smaller brands serve specific needs

Alienware, Gigabyte, Microsoft hold small overall shares, but they serve clearly defined niches: gaming, performance, and the premium productivity segment.

05 Price Distribution

Market Segmentation: Low Budget | Mid-Range | Premium

Price Distribution — Histogram



Segment Definitions

Low Budget

< ₹40.000

216 laptops (21.6%)

Mid-Range

₹40K – ₹80K

468 laptops (46.8%)

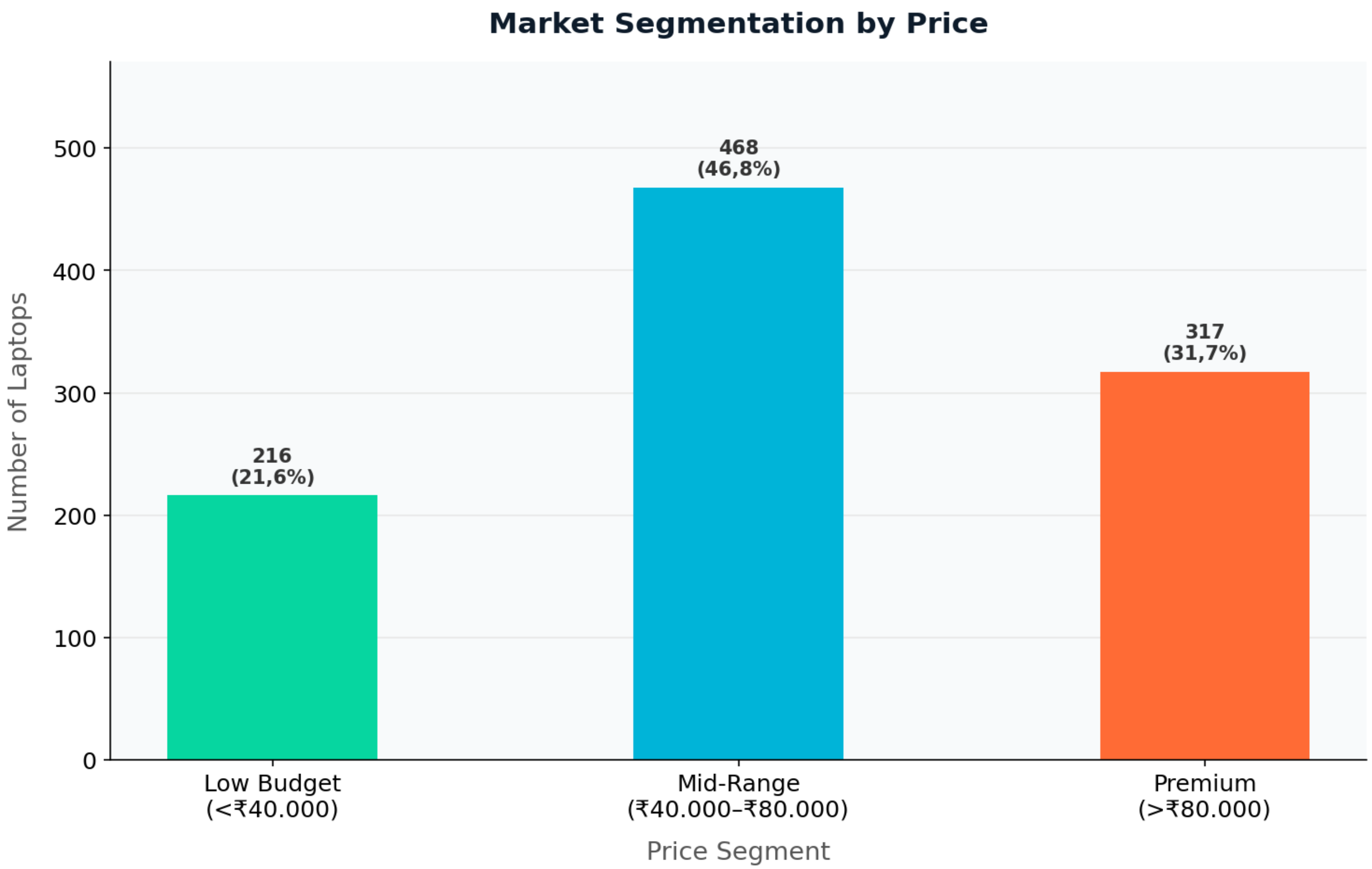
Premium

> ₹80.000

317 laptops (31.7%)

⚠ *Distribution is right-skewed; median ≈ ₹64.990*

Price Distribution — Market Segmentation



Segmentation Insights

Mid-Range segment dominates at 46.8%. Market sweet spot is ₹40K–₹80K.

Premium segment at 31.7% shows high aspirational demand in India.

Low Budget segment at 21.6% is smaller than expected. Entry-level is contracting.

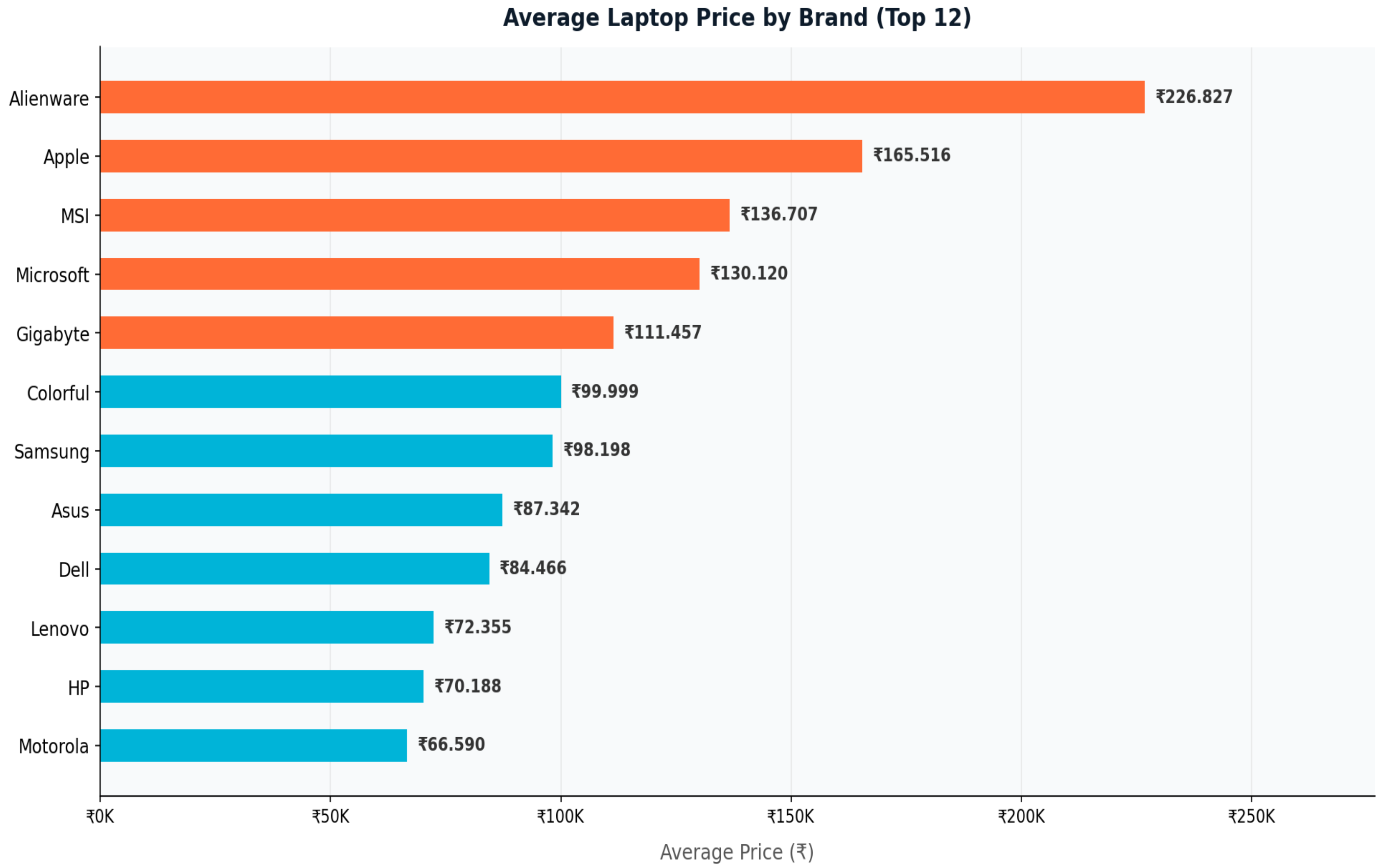
Median (₹64.990) falls in Mid-Range segment. Typical buyer is mid-tier.

Premium segment growth driven by gaming & professional laptop adoption.

06 Brand Positioning

Average Price · Price Spread

Brand Positioning — Average Price by Brand



Brand Tier Classification

Ultra-Premium: >₹100K

Alienware, Apple, MSI, Microsoft

Mid-Premium: ₹60K– ₹100K

Gigabyte, Samsung, Asus, Dell

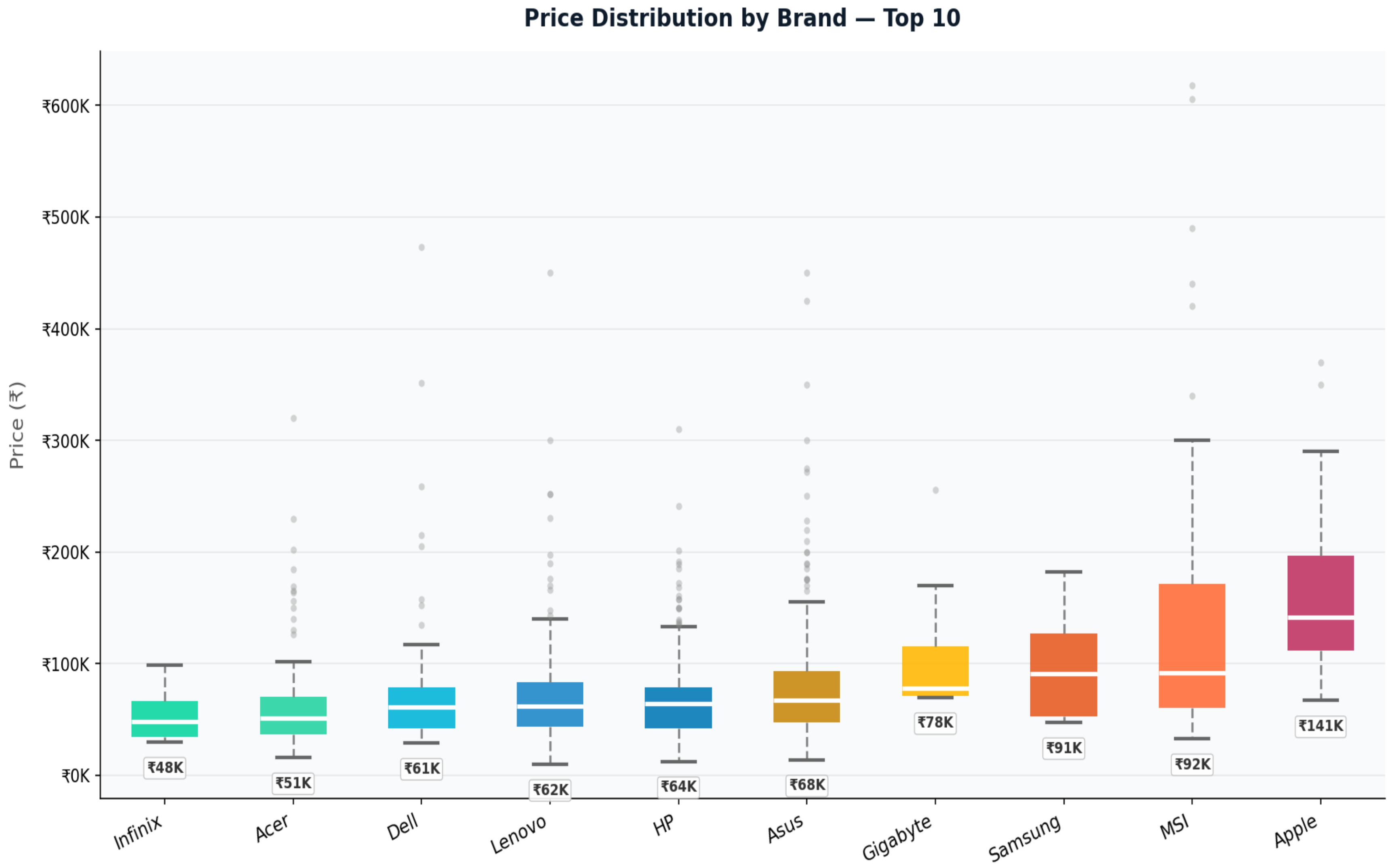
Value Segment: <₹60K

Lenovo, HP, Acer, Infinix

Low Budget Segment: <₹30K

Ultimus, Chuwi, Primebook, AXL

Brand Positioning — Price Spread by Brand



Spread Analysis

Apple has the widest IQR — huge portfolio from ₹80K to ₹300K.

MSI shows consistent premium pricing (tight, high distribution).

HP & Lenovo show broad spreads — cover all segments.

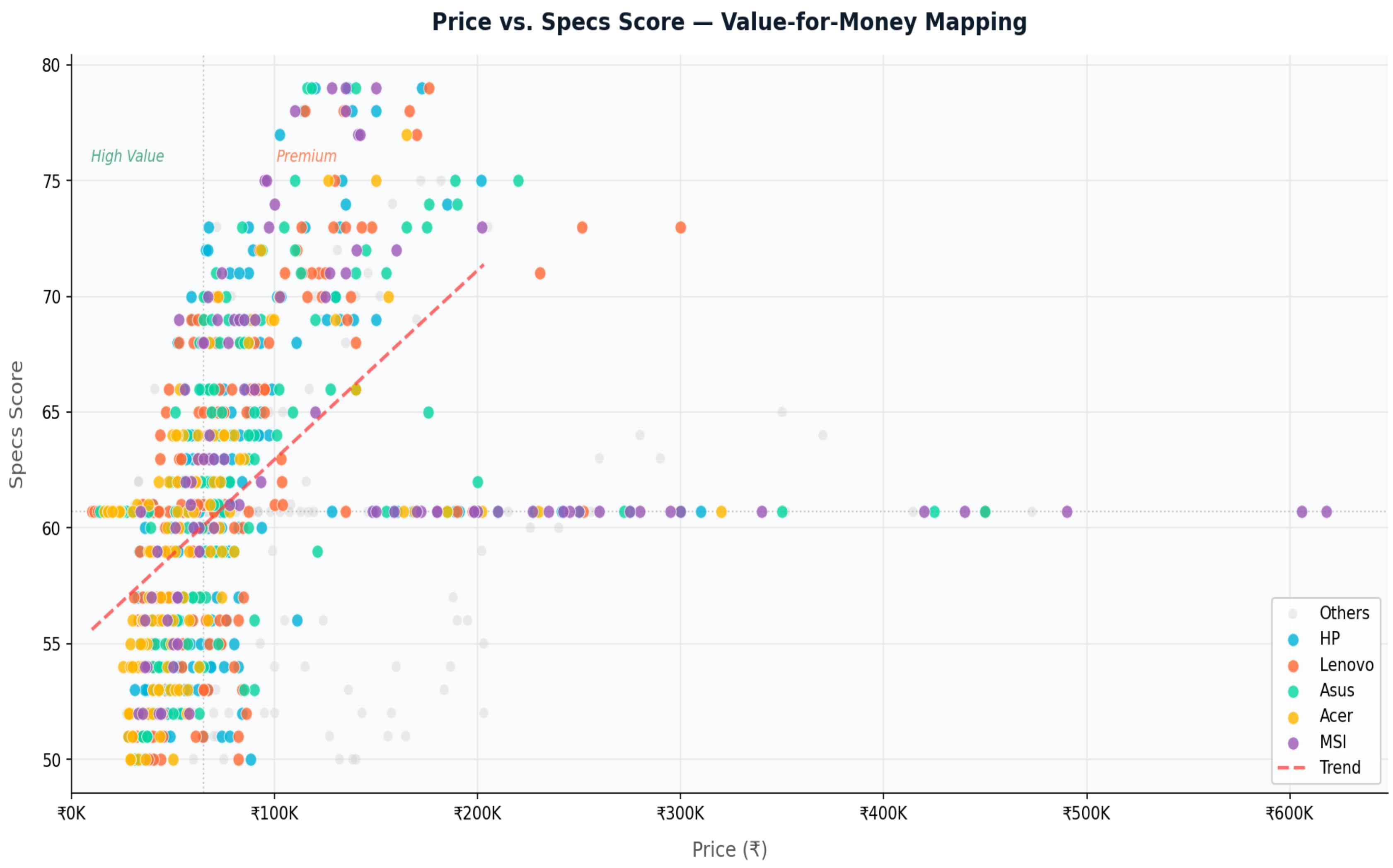
Acer has a compact, lower range — budget-to-mid strategy.

Infinix is the most compact: budget-focused, narrow range.

07 Price vs Performance

Value-for-Money (VFM) Analysis

Price vs Performance — Scatter Analysis



Key Findings

Weak correlation
Higher price ≠ proportionally better specs score (r ≈ 0.4).

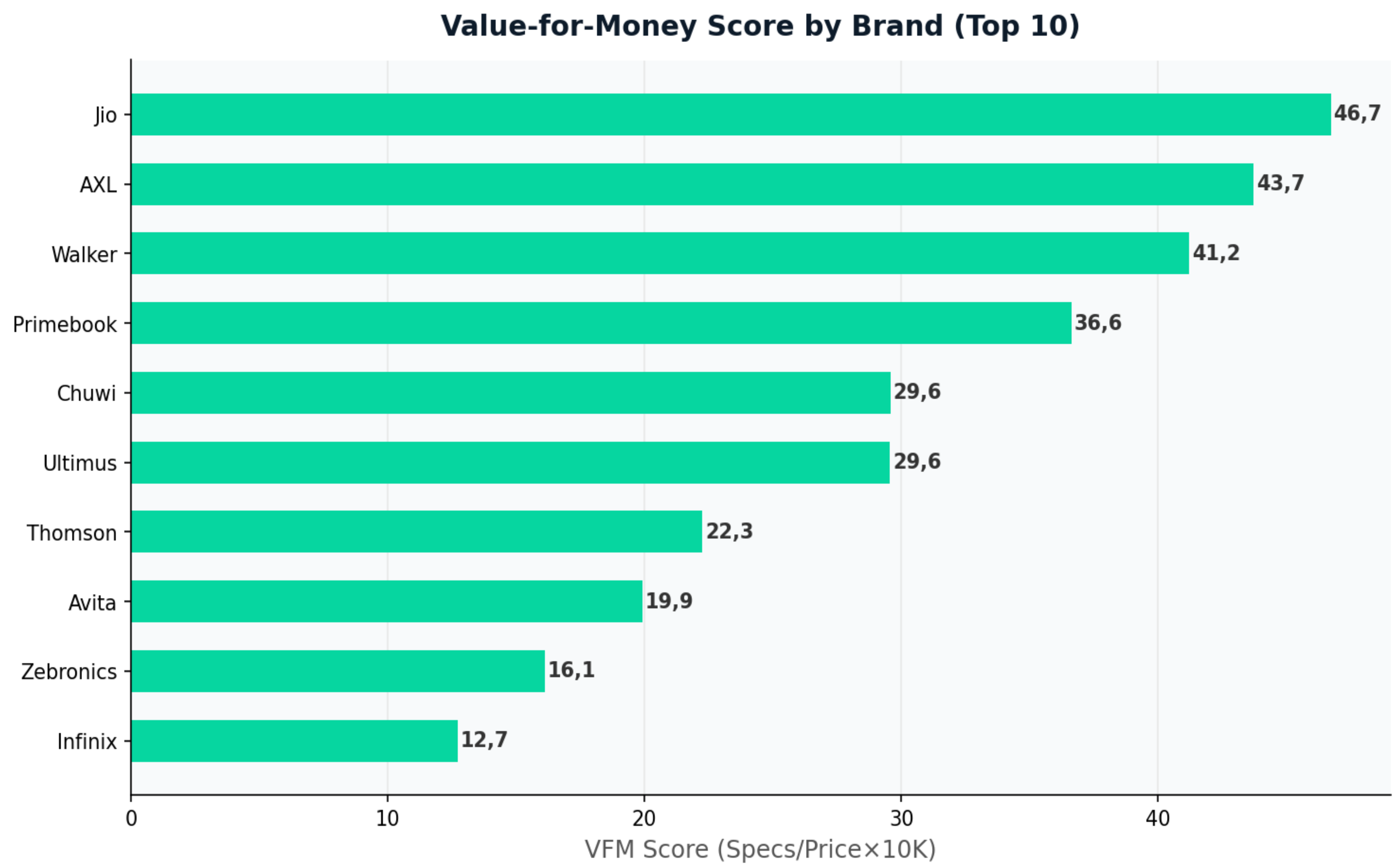
Specs plateau
Scores cluster at 55–70 regardless of price → hardware ceiling effect.

Acer sweet spot
Strong specs at mid-range price; consistent overperformer.

Apple premium gap
Pays for ecosystem/brand, not raw hardware specs.

High Price, Low Value Anomaly
MSI & HP (₹200K – ₹600K) score only 60-61. Premium price reflects brand equity.

Value-for-Money Score by Brand



VFM Methodology

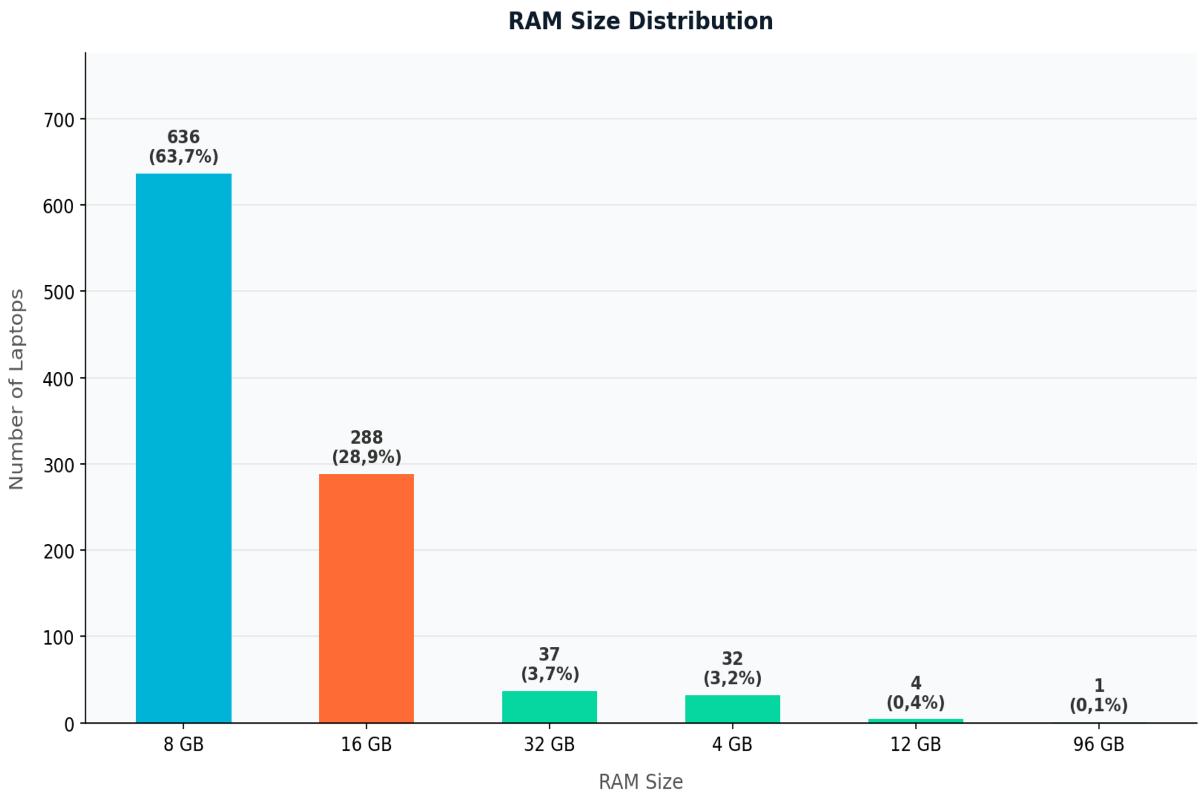
VFM brand = mean [specs_score(i) ÷ (price(i) / ₹10.000)]

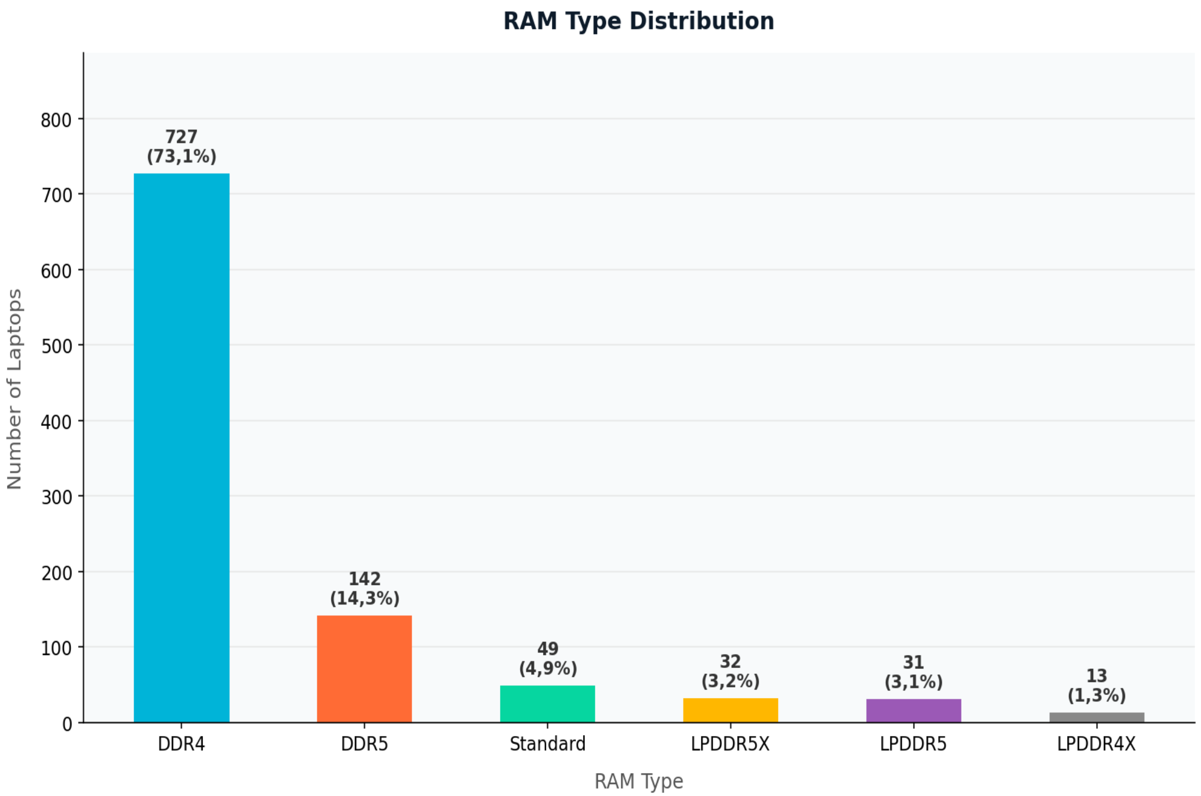
- Low Budget segment brands dominate (Jio, AXL, and Walker) due to extremely low prices.
- Apple (VFM = 4,1) and MSI (VFM= 4,0) lowest VFM score absent from this chart
- VFM ≠ best laptop — also consider build quality, support, longevity.
- Chuwi and Ultimus share the same VFM score despite a ₹1.751 price difference.
- Zebronics has the highest avg specs_score (66,0), yet ranks only 9th in VFM. Better hardware does not guarantee a better VFM rank

08 Hardware Trends

RAM · Storage · Graphics Card

Hardware Trends — RAM Analysis





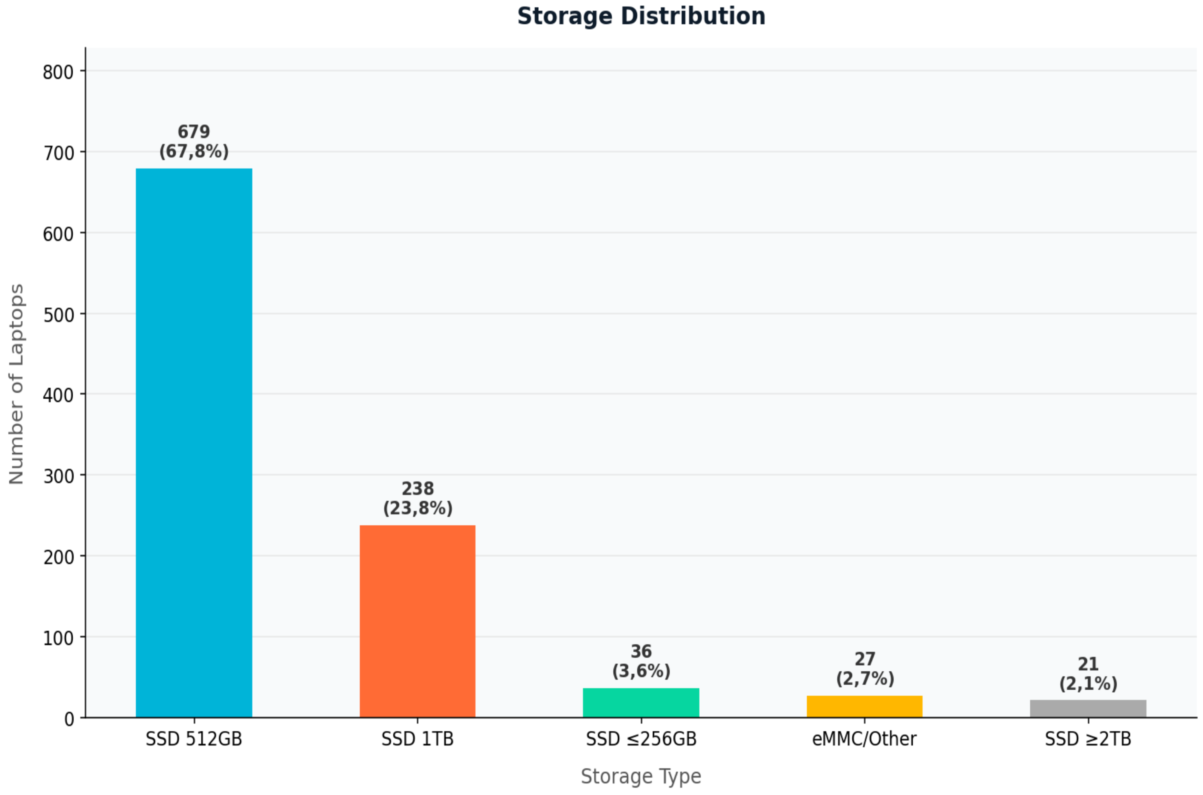
RAM Size Insights

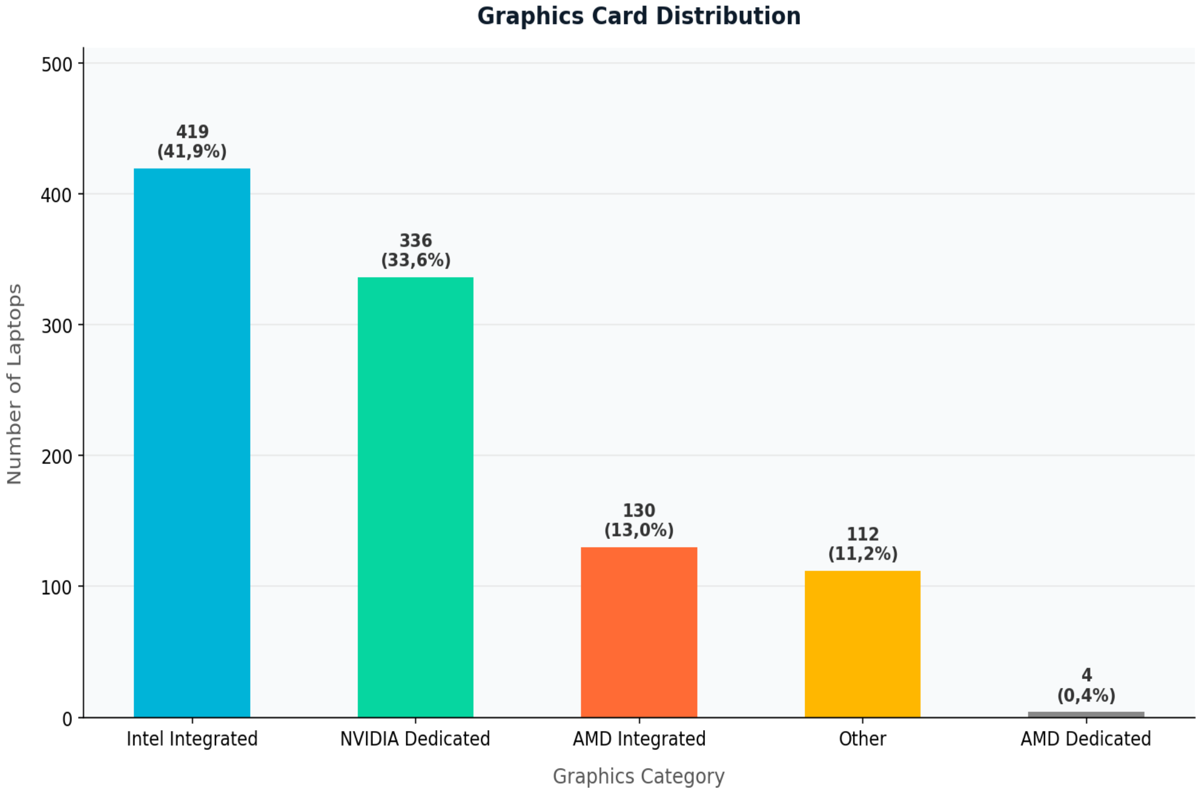
- 8 GB still remains the standard configuration for everyday computing tasks
- 16 GB is gaining around as users demand better multitasking capability.
- 32 GB is still niche, but signals growing adoption among power users.

RAM Type Insights

- DDR4 still dominates because it is reliable, affordable, and works well across all price segments.
- DDR5 appears in 14.3% and is gradually becoming standard in newer, high-end models.
- LPDDR5/5X is found mainly in slim laptops where battery life and portability takes priority over raw performance

Hardware Trends — Storage & Graphics





Storage Insights

- Nearly all laptops now come with SSD storage, making eMMC practically obsolete in today's market.
- 512 GB SSD is the most common choice at 67.8%, striking in the right balance between capacity and affordability.
- 1 TB SSD is picking momentum at 23,8%, driven by users who store more photos, videos, and large files locally.

Graphics Insights

- Most laptops relied on integrated graphics, which is more than enough for everyday office and productivity tasks.
- Dedicated GPUs appear in 34% of laptops, reflecting solid demand from gamers and creative professionals.
- Intel Iris Xe is the most common integrated option, offering a good balance of performance and battery efficiency for mainstream users.

09 Key Insights & Strategic Recommendations

Findings · Recommended Strategic Actions

Findings

Ten findings drawn directly from the data, written to inform decisions rather than just describe numbers.

Five brands own the market.

HP, Lenovo, Asus, Acer, and MSI together hold 82.7% of all product listings. The market is highly consolidated, and any new entrant needs a clear differentiation strategy just to get noticed.

SSD storage is now the standard, not a selling point.

98.1% of laptops in this dataset ship with SSD. eMMC is effectively gone from the mainstream market and only survives in a small number of ultra low-budget devices priced below ₹15K.

The ₹40K to ₹80K range is where the market is won or lost.

46.8% of all laptops fall in this band. It is the single largest segment by product count, and the one where brand battles are most intense and consumer decisions most price-sensitive.

8 GB RAM still leads, but 16 GB is gaining ground fast.

8 GB accounts for 63.5% of the market today, but 16 GB has already grown to 28.8%. A base-spec upgrade cycle is underway, and it is likely to pick up pace as DDR5 becomes more affordable.

A higher price tag does not reliably mean better hardware.

The correlation between price and specs score is around 0.4, which is weak. A significant part of premium pricing in this market is built on brand equity and ecosystem lock-in, not on raw component quality.

Findings (continued)

Findings on value-for-money, premium segments, GPU adoption, brand positioning, and memory trends.

High VFM scores for ultra-budget brands are misleading.

Jio, AXL, and Walker top the VFM chart because their prices are extremely low, not because their hardware is impressive. Among brands with real market reach, Infinix and Acer offer the best specs per rupee.

The Premium segment is bigger than most people expect.

At 31.7%, laptops priced above ₹80K make up almost a third of the market. In a country often described as price-sensitive, this is a meaningful signal that aspirational buying behavior is already here.

Dedicated GPUs are no longer a niche category.

34% of laptops carry a discrete GPU, mostly from NVIDIA RTX. Gaming and content creation are becoming mainstream use cases, and any brand without a strong offering in this space is already falling behind.

Apple commands a price premium the data cannot fully justify.

Apple averages ₹165K, which is 2.5 times the market mean, but its specs score sits comfortably in mid-range territory. Buyers are paying for the ecosystem and the brand experience, not the hardware specifications.

DDR4 is still the mainstream standard, but not for much longer.

DDR5 and LPDDR5 together cover 17.3% of laptops in the dataset. The generational shift is already happening, and DDR4 is likely to move from being the standard to being the budget option within the next product cycle.

Recommended Strategic Actions

Five actions that follow directly from the data, each linked to a specific audience and a specific finding.

For Retailers

Prioritize the mid-range tier in assortment strategy.

With 46.8% of products sitting between ₹40K and ₹80K, this is the market's center of gravity. Retailers who treat it as a secondary category are misreading where consumer demand actually lives.

For Manufacturers

Make 16 GB DDR5 the default before the competitors do.

16 GB RAM already accounts for 28.8% of the market and is growing. The brand that normalises it in mid-range configurations without raising prices significantly will own the upgrade narrative in the next product cycle.

For Consumers

Start looking at Infinix or Acer if value matters most.

Both brands consistently deliver higher specs scores per rupee than their mainstream peers in the ₹40K to ₹65K range. For buyers who want strong hardware without overpaying for a brand name, they are the right starting point.

For Product Strategists

Stop treating gaming laptops as a separate vertical and start building for it.

34% of laptops in this market already carry a dedicated GPU. That is too large a share to manage with a single SKU (Stock Keeping Unit) or a borrowed gaming identity. Brands that build genuine gaming sub-lines now will be harder to displace later.

For Analysts and Future Researchers

The next version of this analysis needs unit sales data.

This study is built on product listings, not actual purchases. That distinction matters. Adding real sales volume, regional data, and year-on-year price history would transform this from a market snapshot into a genuine decision-support tool.

Thank You

Access the Full Analysis

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